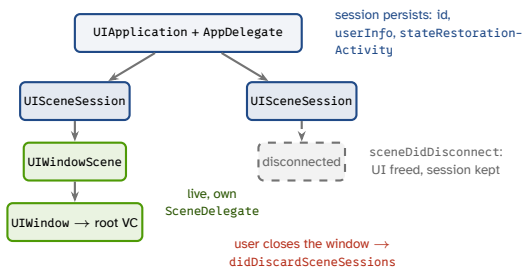
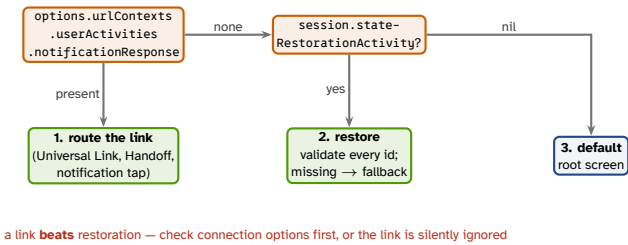


In one line: Since iOS 13 one **process** drives many **UI instances**: each window is a UIWindowScene with its own lifecycle, backed by a persistent UISceneSession. The system may **disconnect** a background scene to save memory and reconnect it later; restoration hands the scene back its own NSUserActivity so it rebuilds **UI state (ids, not model data)** — unless a deep link says where to go.

One process, many scenes



scene(_:willConnectTo:options:) – who decides the first screen?



Scenes and windows

- **AppDelegate** = process: didFinishLaunching, push token, services, configurationForConnecting, didDiscardSceneSessions. **SceneDelegate** = one window: willConnectTo, sceneDidBecomeActive/WillResignActive/WillEnterForeground/DidEnterBackground, sceneDidDisconnect, openURLContexts, continue (links while running). With scenes, the app delegate's applicationDidBecomeActive family is **not called**.
- Info.plist UIApplicationSceneManifest: UIApplicationSupportsMultipleScenes = YES or the app stays single-window on iPad. Open a window with UIApplication.shared.requestSceneSessionActivation(_:userActivity:options:errorHandler:): nil session = new window; an existing one = bring it forward — prefer that to spawning duplicates.
- **Disconnect** ≠ **discard**: disconnect frees a background scene's UI (session kept, may reconnect via willConnectTo); **discard** = the user closed it — clean up per-window data (may arrive at the next launch).
- keyWindow / windows are ambiguous with several scenes: use view.window?.windowScene or connectedScenes.

- Restores after a **system** kill. Discarded after the user closes the window / swipes the app away, or a crash during restore. **Test:** background the app, *then* Stop in Xcode, relaunch.
- Legacy: restorationIdentifier + encodeRestorableState(with:), opt-in shouldSaveSecureApplicationState.
- **SwiftUI:** WindowGroup (multi-instance on iPadOS/macOS); @Environment(\.openWindow) → openWindow(id:) / openWindow(value:). @SceneStorage("k") = small per-scene values, restored automatically; @AppStorage = one UserDefaults for all windows. scenePhase is per scene (combined when read in App).

Example — restore, but let links win

```
func scene(_ scene: UIScene,
           willConnectTo session: UISceneSession,
           options: UIScene.ConnectionOptions) {
    guard let ws = scene as? UIWindowScene else { return }
    window = UIWindow(windowScene: ws)
    if let url = options.urlContexts.first?.url { router.open(url) }
    else if let ua = options.userActivities.first {
        router.continue(ua) } // UL / Handoff
    else if let id = session.stateRestorationActivity?
        .userInfo?["itemID"] as? String, store.exists(id) {
        router.show(id) } // validated
    window?.rootViewController = router.root
    window?.makeKeyAndVisible() }
// asked on entering the background: keep it cheap
func stateRestorationActivity(for scene: UIScene)
    -> NSUserActivity? {
    let ua = NSUserActivity(activityType: "com.you.viewing")
    ua.addUserInfoEntries(from: ["itemID": router.currentID])
    return ua }
```

iPad multitasking

- Split View, Slide Over, Stage Manager (iPadOS 16), freely resizable windows (iPadOS 26), external display: the **system** sizes your window, live. Adapt to **size classes** (`horizontalSizeClass` is `.compact` in a narrow pane on the biggest iPad), never to `userInterfaceIdiom`.
- React to traits: `registerForTraitChanges` (iOS 17; replaces `traitCollectionDidChange`), `viewWillTransition(to:with:)`; `UISplitViewController` / `NavigationSplitView` collapse to a stack in compact width.
- Two windows share the process: a global “current document” breaks. Shared model at app level, per-window state in the scene.

Restoration + SwiftUI

- `userInfo` = property-list values: `ids`, `offsets`, `tab`, `draft text`.
One `NSUserActivity` can also drive Handoff (`isEligibleForHandoff`) and Spotlight.

Traps

- ! Archiving model objects into restoration — store **ids** and re-fetch; the record may be gone.
- ! `@AppStorage` for per-window UI state: two windows clobber each other. `@StateObject` inside `WindowGroup` = one per window.
- ! `NavigationPath` is not `Codable` itself — persist `path.codable` (a `CodableRepresentation`) as `Data`.

Remember

Session persists, scene comes and goes · link > restore > default · ids. not models.